Ratios, Proportions & Percentages

ANSWER KEY

Section 1: Direct & Inverse Proportions

1 If 6 workers can paint a fence in 10 hours, how many hours would it take 15 workers working at the same rate?

a. 4 **hours** ✓

b. 5 hours

c. 6 hours

d. 8 hours

This is an inverse proportion: $\text{Workers} \times \text{Time} = \text{Constant}$. Total worker-hours $= 6 \times 10 = 60$. For 15 workers: $\text{Time} = \frac{60}{15} = 4$ hours.

2 A car travels 240 km using 20 liters of fuel. How many liters are required to travel 360 km at the same rate?

a. 30 **liters** ✓

b. 28 liters

c. 32 liters

d. 35 liters

Direct proportion: $\frac{240}{20} = 12$ km/L. For 360 km: Fuel $= \frac{360}{12} = 30$ liters.

3 If $a : b = 2 : 3$ and $b : c = 4 : 5$, what is the ratio $a : c$?

a. $8 : 15$ ✓

b. $2 : 5$

c. $6 : 15$

d. $8 : 12$

Equate b by multiplying: $a : b = 8 : 12$ and $b : c = 12 : 15$. Thus, $a : c = 8 : 15$.

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Section 2: Percentages, Discounts & Markups

4 A jacket originally priced at 150 SAR is discounted by 20%. What is the sale price?

a. 120 **SAR** ✓

b. 110 SAR

c. 125 SAR

d. 130 SAR

Discount $= 0.20 \times 150 = 30$ SAR. Sale price $= 150 - 30 = 120$ SAR.

5 The price of an item was increased by 20% and then decreased by 20%. What is the net percentage change in price?

a. 4% decrease ✓

b. 0%

c. 2% decrease

d. 4% increase

Net multiplier $= (1 + 0.20)(1 - 0.20) = 1.20 \times 0.80 = 0.96$. A final factor of 0.96 represents a 4% net decrease.

6 If 25% of a number x is 45, what is 60% of x?

a. 108 ✓

b. 90

c. 120

d. 135

If $0.25x = 45$, then $x = \frac{45}{0.25} = 180$. Then 60% of $180 = 0.60 \times 180 = 108$.

Section 3: Speed, Distance & Average Velocity

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7 A car travels from City A to City B at 60 km/h and returns along the same route at 40 km/h. What is the average speed for the round trip?

a. 48 km/h ✓

b. 50 km/h

c. 45 km/h

d. 52 km/h

Harmonic mean formula for equal distances: $v_{\text{avg}} = \frac{2v_1v_2}{v_1+v_2} = \frac{2(60)(40)}{60+40} = \frac{4800}{100} = 48$ km/h.

8 Two trains leave the same station in opposite directions at speeds of 70 km/h and 80 km/h. In how many hours will they be 450 km apart?

a. 3 **hours** ✓

b. 2.5 hours

c. 3.5 hours

d. 4 hours

Relative speed in opposite directions $= 70 + 80 = 150$ km/h. Time $= \frac{\text{Distance}}{\text{Relative Speed}} = \frac{450}{150} = 3$ hours.

9 A cyclist travels 15 km in 45 minutes. What is their speed in kilometers per hour?

a. 20 km/h ✓

b. 18 km/h

c. 25 km/h

d. 15 km/h

45 minutes $= \frac{45}{60} = \frac{3}{4}$ hour. Speed $= \frac{15}{3/4} = 15 \times \frac{4}{3} = 20$ km/h.

Section 4: Work Rates & Shared Labor

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10 Pipe A can fill a tank in 6 hours, while Pipe B can fill it in 3 hours. If both pipes are opened together, how long will it take to fill the tank?

a. 2 **hours ✓**

b. 2.5 hours

c. 1.5 hours

d. 3 hours

Combined rate $= \frac{1}{6} + \frac{1}{3} = \frac{1}{6} + \frac{2}{6} = \frac{3}{6} = \frac{1}{2}$ tank/hour. Time required $= 2$ hours.

11 Ahmed can complete a job in 4 days, and Khalid can complete it in 6 days. If they work together, how many days will the job take?

a. 2.4 **days ✓**

b. 2.5 days

c. 3 days

d. 2 days

Combined rate $= \frac{1}{4} + \frac{1}{6} = \frac{3+2}{12} = \frac{5}{12}$. Time $= \frac{12}{5} = 2.4$ days.

12 A tank has a leak that can empty it in 12 hours. If an inlet pipe fills it in 4 hours, how long will it take to fill the tank with the leak open?

a. 6 **hours ✓**

b. 5 hours

c. 8 hours

d. 4.5 hours

Net rate $= \frac{1}{4} - \frac{1}{12} = \frac{3-1}{12} = \frac{2}{12} = \frac{1}{6}$. Time required $= 6$ hours.

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